

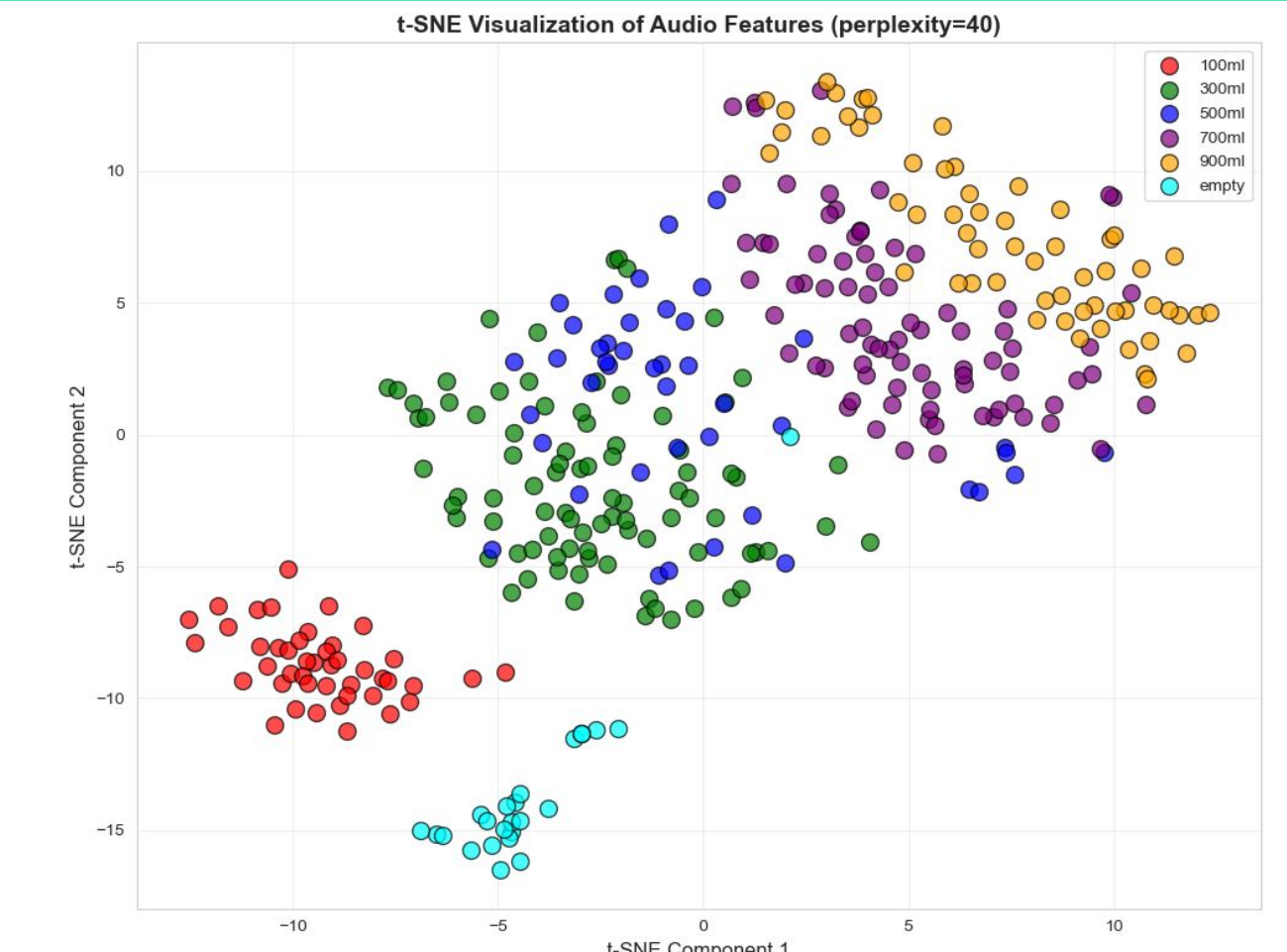
HydroSense

Saideep Kasipathy, Yunjong (Kevin) Na, Min Park
{saideepk, kevinyjn, phyunmin}@umich.edu

Introduction / Motivation: Can You Tell Water Level Just by Shaking a Bottle?

We set out to build a low-cost, simple water-level tracker that could work universally across different non-transparent water bottles. The goal was to give people an easy way to track their water intake, regardless of the bottle they're using and without needing any expensive or specialized equipment.

- Prove a SVM is effective (hopefully small enough to run on-board in the future)
- Affordable VPU (bone conduction microphone robust to noises in the environment)
- MCU that processes audio data efficiently



Technical Challenges

- LED timing + DMA configuration - Getting the Neopixel LED strip working required debugging conflicting online guides; on our STM32 we ultimately had to configure the timer's DMA as memory-to-peripheral with 32-bit word data width (not half word), which we found only through experimentation.
- Another challenge was configuring the VPU. Because of the lack of documentation on how to configure DFSDM on the STM32 board series it took longer than expected for us to understand the filter and find an optimal oversampling ratio for pdm → pcm conversion with accurate resolution. We were able to fix this issue through number of trial and errors.

How HydroSense Works

